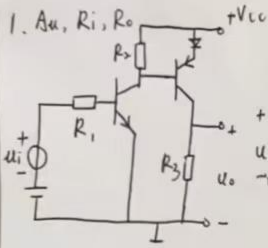
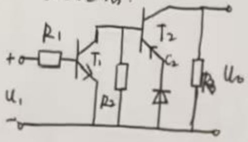
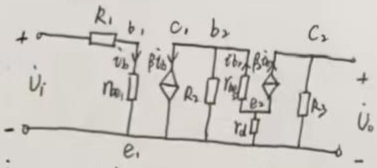
1. A_u, R_i, R_o



交流通路:



微变等效电路:



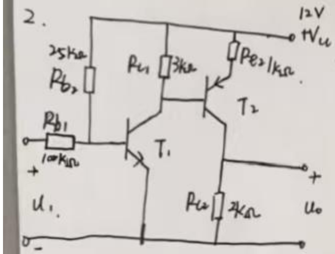
$$r_d=\frac{nU_T}{I_{DQ}}$$

$$\dot{A}_{u1}=-\frac{\beta_1\{R_2//[r_{be2}+(1+\beta_2)r_d]\}}{r_{be}+R_1}\qquad \dot{A}_{u2}=-\frac{\beta_2R_3}{r_{b2}+(1+\beta_2)r_d}$$

$$\dot{A}_u=\dot{A}_{u1}\cdot\dot{A}_{u2}=-\frac{\beta_1\{R_2//[r_{b2}+(1+\beta_2)r_d]\}}{r_{be}+R_1}\cdot\left[\frac{R_2R_3}{r_{be2}+(1+\beta_2)r_d}\right]$$

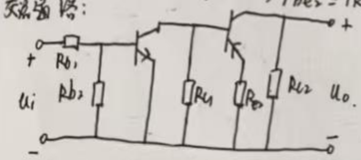
$$R_i=R_1+r_{be1}\qquad R_o=R_3.$$

2.

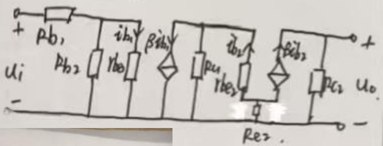


$\beta_1=80, r_{be1}=1k\Omega, \beta_2=60, r_{be2}=1k\Omega$. A_u, R_i, R_o.

交流通路:



微变等效电路:



$$\dot{A}_{u1}=-\frac{\beta_1\cdot i_{b1}\cdot\{R_{c1}//[r_{be2}+(1+\beta_2)R_{e2}]\}}{r_{be1}\cdot i_{b1}}=-\frac{\beta_1\{R_{c1}//[r_{be2}+(1+\beta_2)R_{e2}]\}}{r_{be1}}=$$

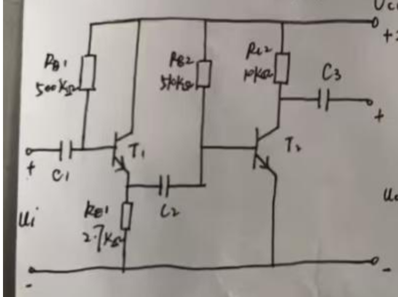
$$\dot{A}_{u2}=-\frac{\beta_2\cdot i_{b2}\cdot R_{c2}}{i_{b2}\cdot[r_{b2}+(1+\beta_2)R_{e2}]}=-\frac{\beta_2R_{c2}}{r_{b2}+(1+\beta_2)R_{e2}}=$$

$$\dot{A}_u=\dot{A}_{u1}\cdot\dot{A}_{u2}=$$

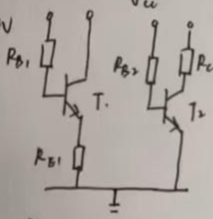
$$R_i=R_{b1}+(R_{b2}//r_{be1})=$$

$$R_o=R_{C2}=$$

3. $\beta_1=\beta_2=50$. A_u, R_i, R_o.



直流通路:



$$i_{B1}=\frac{V_{CC}-U_{BEQ1}}{R_{B1}+(1+\beta)R_E}=36.5\mu A$$

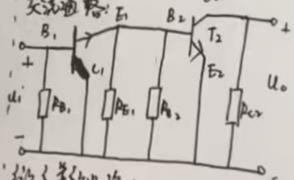
$$i_{B2}=\frac{V_{CC}-U_{CEQ2}}{R_{B2}}=45.7\mu A$$

$$I_{EQ}=(1+\beta)I_{BQ}$$

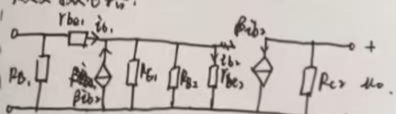
$$r_{be1}=300\Omega+\frac{26mV}{I_{EQ}}(1+\beta)=1k\Omega$$

$$r_{be2}=868\Omega.$$

交流通路:



微变等效电路:



$$\dot{A}_{u1}=\frac{(1+\beta)i_{b1}\cdot(R_{E1}//R_{B2}//r_{b2})}{(1+\beta)i_{b1}(R_{E1}//R_{B2}//r_{b2})+i_{b1}\cdot r_{b1}}$$

$$\dot{A}_{u2}=-\frac{\beta_2\cdot i_{b2}\cdot R_{C2}}{i_{b2}\cdot r_{be2}}$$

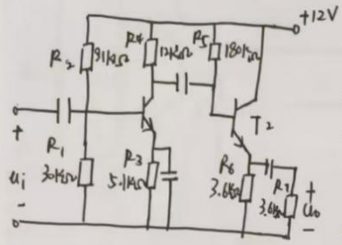
$$\cancel{A_u}=\dot{A}_u=\dot{A}_{u1}\cdot\dot{A}_{u2}$$

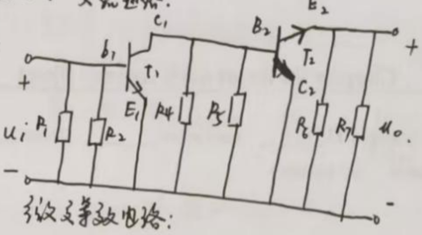
$$R_i=R_{B1}//[r_{be1}+(1+\beta_1)(R_{E1}//R_{B2}//r_{be2})]$$

$$R_o=R_{C2}$$

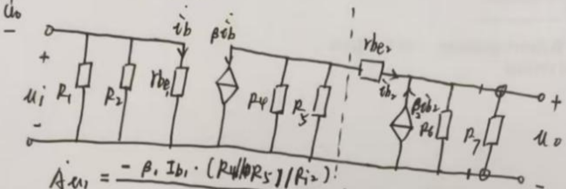
4. $\beta_1=\beta_2=100$, $r_{be1}=6K\Omega$, $r_{be2}=1.5K\Omega$、 交流通路:

A_u, R_i, R_o.





微变等效电路:



$$\dot{A}_{u1}=\frac{-\beta_1 I_{b1}\cdot(R_4/\!/R_5/\!/R_{i2})}{I_{b1}\cdot r_{be1}}$$

$$R_{i2}=r_{be2}+(1+\beta_2)(R_6/\!/R_7).$$

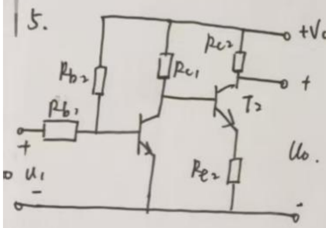
$$\dot{A}_{u2}=\frac{(1+\beta_2)(R_6/\!/R_7)I_{b2}}{(1+\beta_2)(R_6/\!/R_7)I_{b2}+r_{be2}\cdot I_{b2}}$$

$$\dot{A}_u=\dot{A}_{u1}\cdot\dot{A}_{u2}$$

$$R_i=R_1/\!/R_2/\!/r_{be1}$$

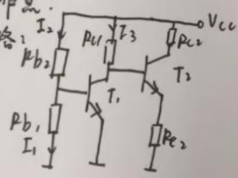
$\cancel{R_o=R_5/\!/R_6/\!/\frac{r_{be2}}{1+\beta_2}}$. $R_o=R_6/\!/\frac{R_4/\!/R_5+r_{be2}}{1+\beta_2}$

5.



静态工作点:

直流通路:



$V_{cc}=\cancel{I_{Rb}}\,I_2\cdot R_{b2}+I_1\cdot R_{b1}$

$V_{cc}=I_2\cdot R_{b2}+U_{BEQ1}$

$\Rightarrow I_1=\frac{U_{BEQ1}}{R_{b1}}$ $\quad I_2=\frac{V_{cc}-U_{BEQ1}}{R_{b2}}$

$I_{bQ1}=I_2-I_1=\frac{V_{cc}-U_{BEQ1}}{R_{b2}}-\frac{U_{BEQ1}}{R_{b1}}.$

$I_{cQ1}=\beta_1 I_{b1}=\beta_1\left(\frac{V_{cc}-U_{BEQ1}}{R_{b2}}-\frac{U_{BEQ1}}{R_{b1}}\right).$

$I_3=I_{cQ1}+I_{bQ2}$

$V_{cc}=I_3R_{c1}+U_{BEQ2}+(1+\beta_2)I_{bQ2}\cdot R_{e2}$

$\Rightarrow I_{bQ2}=I_3-I_{c1}$ $\quad\Rightarrow I_3=\frac{V_{cc}-V_{BEQ2}-(1+\beta_2)I_{bQ2}R_{e2}}{R_{c1}}.$

$\Rightarrow V_{cc}=I_3\cdot R_{c1}+U_{CEQ1}$

$V_{cc}=I_{c2}R_{c2}+U_{CEQ2}+I_{e2}\cdot R_{e2}$

$I_{e2}=(1+\beta_2)I_{bQ2}$, $\quad I_{cQ2}=\beta_2 I_{bQ2}$

可得: $I_{BQ1}=\frac{V_{cc}-U_{BEQ1}}{R_{b2}}-\frac{U_{BEQ1}}{R_{b1}}$

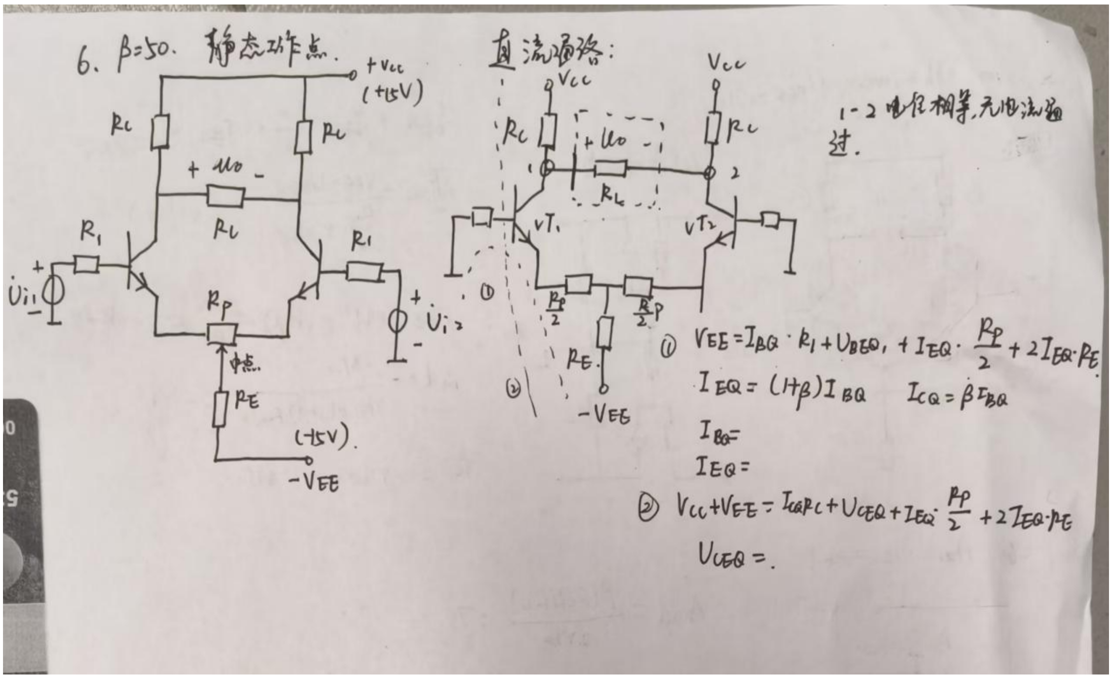
$I_{CQ1}=\beta_1\left(\frac{V_{cc}-U_{BEQ1}}{R_{b2}}-\frac{U_{BEQ1}}{R_{b1}}\right)$

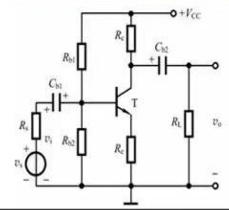
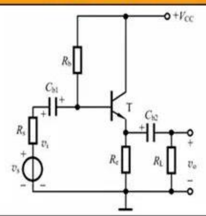
$U_{CEQ1}=V_{cc}-R_{c1}\cdot I_3=\cdots$

$I_{BQ2}=I_3-I_{CQ1}=\cdots$

$I_{CQ2}=\beta_2 I_{BQ2}=\cdots$

$U_{CEQ2}=V_{cc}-\beta_2 I_{BQ2}R_{c2}-(1+\beta_2)I_{BQ2}\cdot R_e=\cdots$



	共射放大电路	共集放大电路
电路结构		
电压放大倍数	$A_u = -\dfrac{\beta R_L'}{r_{be} + (1+\beta)R_e}$	$A_u = \dfrac{(1+\beta)R_L'}{r_{be} + (1+\beta)R_L'}$
电流放大倍数	$A_i \approx \beta$	$A_i \approx 1+\beta$
输入、输出电压相位关系	反相	同相
输入电阻	$R_i = R_{b1} // R_{b2} // [r_{be} + (1+\beta)R_e]$	$R_i = R_b // r_{be} + (1+\beta)R_L'$
输出电阻	$R_o = R_C$	$R_o = R_E // \dfrac{r_{be} + R_s'}{1+\beta} \approx \dfrac{r_{be} + R_s'}{1+\beta}$